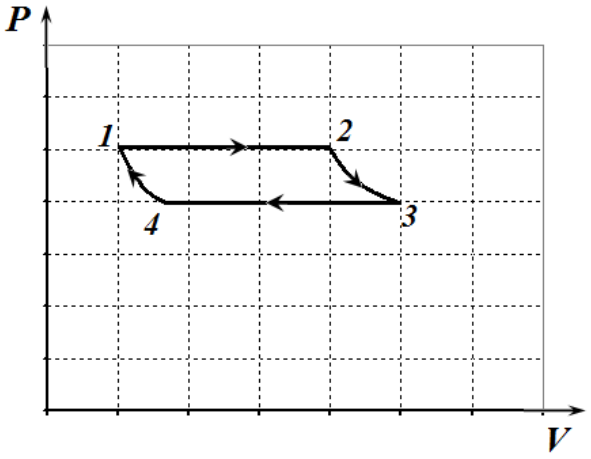
이론시험 문제
해설
문제 1 (10점)
문제 1A (3점)

1
.
3
.
2·4 .
(***P,V***)
.



, .

$$\eta = \frac{T_1 - T_2}{T_1}, \qquad (1)$$

T_1 1
, T_2 3 .

.
1

$$P_1 = P_0 + \frac{(M+m)g}{S} \approx 1,3 \cdot 10^5 Pa\,. \qquad (2)$$

,

$$t_1 = \frac{P_1 + b}{a} = \frac{130 + 384}{4,85} \approx 106°C = 379K\,. \qquad (3)$$

(1)

$$T_1 - T_2 = \frac{P_1 - P_2}{a} = \frac{mg}{Sa} = \frac{20}{4,85} \approx 4,2\,K\,. \qquad (4)$$

$$\eta = \frac{T_1 - T_2}{T_1} = \frac{4,2}{379} = 1,1\%\,.$$

채점 기준

№	평가 항목	배점
1.	1 3 ·	0,25
2.	2 4	0,25
3.		1,0
4.		0,75
5.	(1)	0,25
6.	(2) (4)	0,25
7.		0,25
합계		**3.0**